

# 4-bit UP/DOWN Counter using SystemVerilog Enum

Minor Project - Design, Simulation and Verification

| Item                  | Specification                                                         |
|-----------------------|-----------------------------------------------------------------------|
| HDL                   | SystemVerilog                                                         |
| Counter width         | 4 bits                                                                |
| Direction             | UP / DOWN                                                             |
| Special behavior      | Value 4 appears for two consecutive clock cycles during DOWN counting |
| Reset                 | Synchronous, active-high                                              |
| Recommended simulator | EDA Playground                                                        |

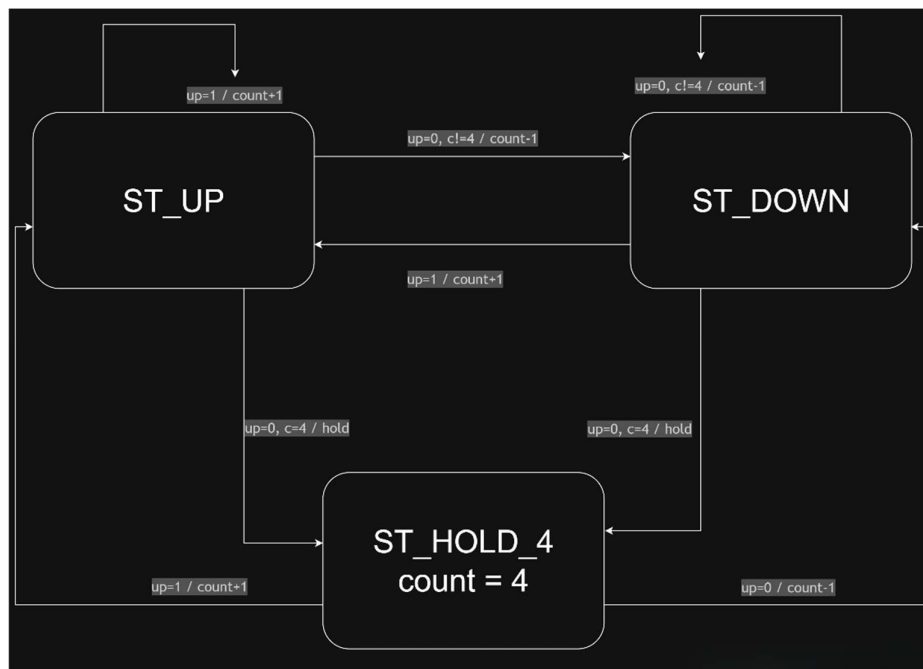
## Problem Definition

Simulate the UPDOWN counter counting 4 twice while down count using SV (enum datatype)

## Design Approach

The counter holds a 4-bit count value. A SystemVerilog enum type holds the operational state, with three possible values: ST\_UP, ST\_DOWN, and ST\_HOLD\_4. When counting DOWN and the count reaches 4, the state is changed to ST\_HOLD\_4. The count value is held for the next clock cycle, and then changed back to ST\_DOWN to produce 3.

| State     | Meaning                              |
|-----------|--------------------------------------|
| ST_UP     | Normal upward counting               |
| ST_DOWN   | Normal downward counting             |
| ST_HOLD_4 | Hold the value 4 for one clock cycle |



## Functional Behavior

UP counting:

0 -> 1 -> 2 -> 3 -> 4 -> 5 -> 6 -> ... -> 15 -> 0 DOWN counting (with the special case behavior)

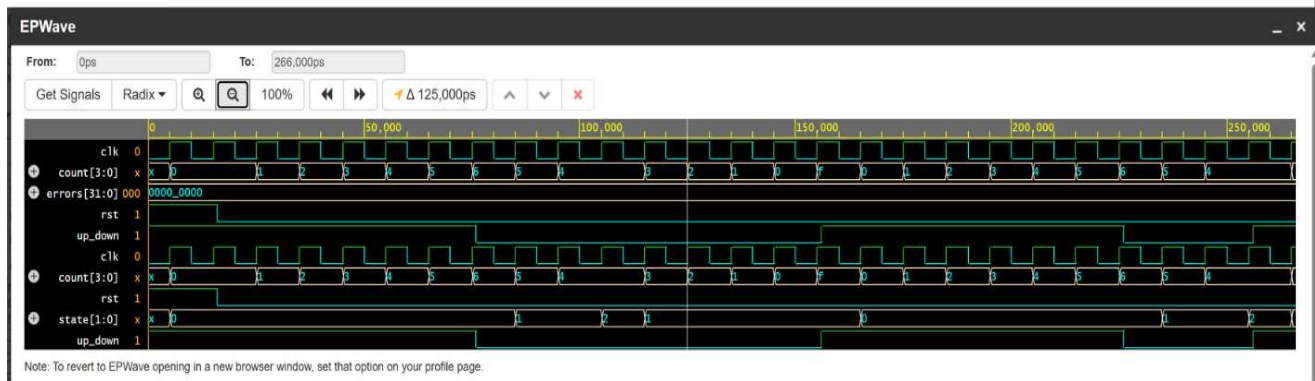
6 -> 5 -> 4 -> 4 -> 3 -> 2 -> 1 -> 0 -> 15 -> ...

The count value is updated on the rising edge of the clock. The synchronous reset sets the count to zero...

## Verification

The testbench creates a 10 ns clock, resets the counter, tests the UP counting, changes the direction to DOWN, and verifies the special case behavior. It also verifies the transition from 0 to 15 when counting DOWN. Each of the expected values is compared to the actual output and marked as PASS or FAIL.

| Test          | Expected result                 |
|---------------|---------------------------------|
| Reset         | count = 0                       |
| UP sequence   | 0 -> 1 -> 2 -> 3 -> 4 -> 5 -> 6 |
| Double-4      | 6 -> 5 -> 4 -> 4 -> 3           |
| Continue DOWN | 3 -> 2 -> 1 -> 0                |
| DOWN wrap     | 0 -> 15                         |



## #output

[2026-09-04 15:15:58 UTC] iverilog '-Wall' '-g2012' design.sv testbench.sv && unbuffer vvp a.out

VCD info: dumpfile dump.vcd opened for output.

=== 4-bit UP/DOWN Counter Test ===

PASS: reset | count=0

PASS: UP 0->1 | count=1

PASS: UP 1->2 | count=2

PASS: UP 2->3 | count=3

PASS: UP 3->4 | count=4

PASS: UP 4->5 | count=5

PASS: UP 5->6 | count=6

PASS: DOWN 6->5 | count=5

PASS: DOWN 5->4 (first 4) | count=4

PASS: DOWN 4->4 (second 4) | count=4

PASS: DOWN 4->3 | count=3

PASS: DOWN 3->2 | count=2

PASS: DOWN 2->1 | count=1

PASS: DOWN 1->0 | count=0

PASS: DOWN 0->15 wrap | count=15  
PASS: UP back to 6 | count=6  
PASS: DOWN 6->5 (again) | count=5  
PASS: first 4 (again) | count=4  
PASS: second 4 (in ST\_HOLD\_4) | count=4  
PASS: HOLD -> UP 4->5 | count=5  
=== ALL 20 CHECKS PASSED ===  
testbench.sv:81: \$finish called at 266000 (1ps)  
Finding VCD file...  
./dump.vcd  
[2026-09-04 15:15:59 UTC] Opening EPWave...  
Done

## Software

**Recommended:** EDA Playground with a SystemVerilog simulator. ModelSim/Questa or Vivado can also be used. Only simulation is needed, no development boards required.

## FilesFiles

|                      |               |
|----------------------|---------------|
| updown_counter.sv    | RTL design    |
| updown_counter_tb.sv | Testbench     |
| README.md            | Project notes |
| report.pdf           | This report   |